Numpy → Library → Scientific computation

Provide → multidimensional array object

- mathmatical
- logical ✓
- shape manipulation ✓
- sorting
- selection
- I/O (Data Read/write) ✓

numpy 2? Data type = ndarray object

✓ python = List

Homogeneous Data types
↓
same

why we need numpy :

2008 < ML , Data science
↓
MatLab , R programming

python → software development
↓
consider slow

Big Data

- List , Dic

→ List, Dic

Then someone make

Numpy Package

New Data type → numpy array

C programming data type

wrapper / Layer

Python (used)

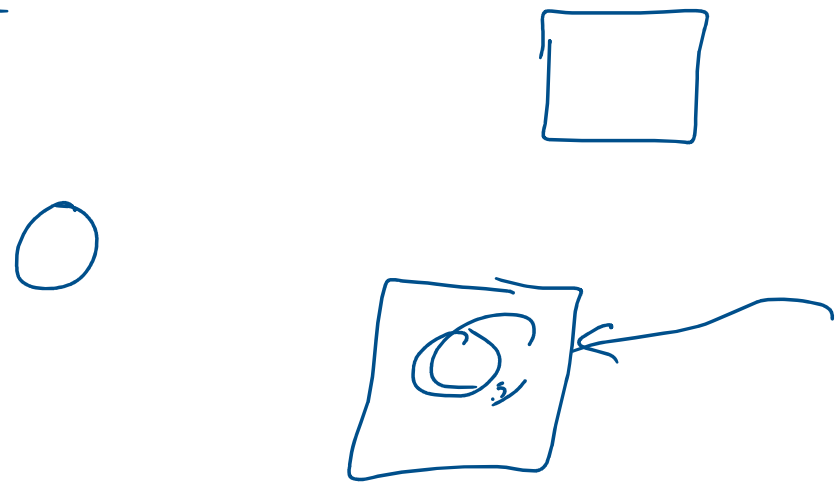
Python → syntax → simple, powerful, easy to learn

Numpy Library → python (ML, DS)

↳ Base

pandas, matplotlib, scik-learn

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$



int a[5]

[0, 0, ..., 0]

-3 -2 -1

0 1 2

-3 0

-2 1

-1 2

```
[[10 20 30]
 [40 50 60]
 [70 80 90]]
```

a[1][1]

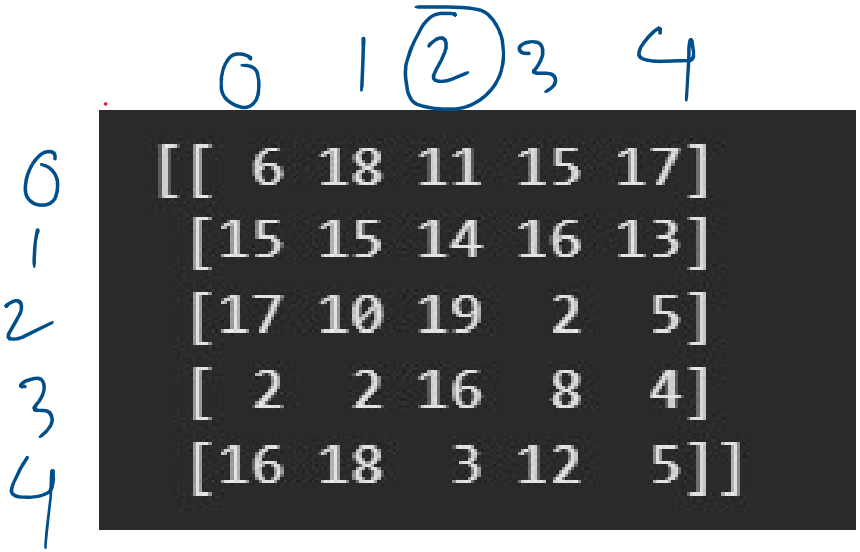
[30, 60, 90]

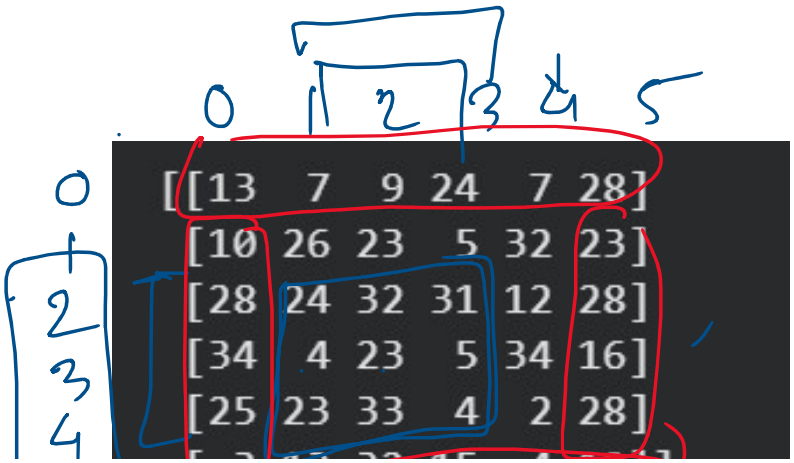
[ ] [[10, 20]
[40, 50]]

10 20 30 40 50 60

(5, 5)

(5, 5)





3, 5 row

2, 4 col

```
[25 23 33  4  2 28]
[ 3 13 32 15  4 19]]
```

```
[[13  7  9 24  7 28]
 [10 26 23  5 32 23]
 [28 24 32 31 12 28]
 [34  4 23  5 34 16]
 [25 23 33  4  2 28]
 [ 3 13 32 15  4 19]]
```

a, b, c, d

[ ] →

c = [ ] = 0

a[:, :]

```
[[13  7  9 24  7 28]
 [10 26 23  5 32 23]
 [28 24 32 31 12 28]
 [34  4 23  5 34 16]
 [25 23 33  4  2 28]
 [ 3 13 32 15  4 19]]
```

```
[[1 1 1 1]
 [2 2 2 2]
```

[4, 8, 6, 5] → row

```
[[1 1 1 1]
 [2 2 2 2]
 [2 2 1 1]
 [1 1 1 2]]
```

[6, 6, 5, 6]

1, 2, 3, 4, 5, 6, 7, 8

[1, 2, 3, 4 ..]

(1, 4)

[[1, 2, 3]
  [4, 5, 6]]

(1, 6) (2, 3)

(6, 1) (3, 2)

[[4, 5, 6]]

[[1, 2, 3, 4, 5, 6]]

6 ! =

[1, 2, 3]

3x3

$$\begin{matrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$